

This document is meant to be used in the following way. Each chapter is a brief recap of what was in the slides. **Take Home Concepts** is a list of concepts recapped at the end of the slides while **Stuff to Know** is what is listed under the # Open-ended exam questions. The **Quiz** part contains the quizzes he uploaded with solutions and explanations.

1) Acoustics and Psychoacoustics

ANSI/ASA describes acoustic as **science of sound including production, transmission and effects, including biological and psicological effects. The qualities of a room together determine the character wrt adutiroy effects**

1.1) Acoustics

The wave equation is:

$$\frac{\partial^2 p}{\partial x^2} - \frac{1}{c^2} \frac{\partial p}{\partial t^2} = 0$$

where p is the pressure and c the speed of sound.

The general solution (with c constant) is a **longitudinal** wave

$$p = f(ct - x) + g(ct + x)$$

with f, g twice differentiable functions

Recall the fundamental properties of waves:

$$\lambda = \frac{v}{f} \quad f = \frac{1}{T}$$

Each periodic signal can be decomposed into a series of simple periodi cwaves called harmonics. The set of all harmonics is called the spectrum:

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^N a_n \cos(nt) + b_n \sin(nt)$$

For periodic waves, the partials are integer multiples of the fundamental frequency f_0 .

A standing wave is a wave where the peaks (max/min) and nodes (zero) do not change position over time.

For auditory communications we have:

- Source: vibrating object with energy injected to it
- Medium: allows for sound propagation
- Receiver: ear-brain transforms acoustic pressure into neural impulses

The warmer a medium the faster the sound. Steel has the fastest propagation. It propagates the fastest in solids, then liquids then gasses. In air at 0 C it is 331.5 m/s, at 20 C it is 344 m/s

Tissue	Velocity of Sound (m/sec)	Substance	Temperature (°C)	Speed (m/sec)
Air	330	CO ₂	0	258
Fat	1450	CO ₂	35	274
Water	1480	Air	0	331.5
Average Human Soft Tissue	1540	Air	20	344
Brain	1540	Water Vapor	35	402
Liver	1550	Helium	20	927
Kidney	1560	Hydrogen	0	1,270
Blood	1570	Water	15	1,437
Muscle	1580	Steel	-	5,000
Lens of Eye	1620			
Skull Bone	4080			

Table

There are three domains of acoustics based on where the sound lives: Air, ear, brain:

Air	Ear	Brain
Sound	Sensation	Music, words, ...
Amplitude/Intensity	Loudness	Dynamics
Frequency	Pitch	Tone classes
Spectrum	Timbre	Musical instruments
Radiation	Localization	Subjective spatial map

3 domains

The brain might trick us:

- Shepard tones: seemingly infinite increase in pitch
- Aural harmonics as integer multiple of original tone
- Tartini's third note: play a perfect fifth, the brain will hear a third note one octave below (perfect fifth is $f_2 = \frac{3}{2} f_1 \rightarrow \Delta f = \frac{1}{2} f_1$)
- Missing Fundamental: If the fundamental is removed from a harmonic spectrum, it is heard all the same

Beats are a **amplitude modulation** phenomenon that allows two similar tone to be perceived as one beating tone. The beat frequency is $f_b = \Delta f$

1.2) Auditory System TODO

The ear is divided into external, middle and inner ear.

The inner ear contains the **cochlea**

The **basilar membrane** is **narrow-rigid at the base, wide-soft at the apex**. This is a bank of filters tuned at specific vibrations.

Low at base, high at apex.

It is non-linear $\rightarrow$ additional aural harmonics

1.3) Elements of Psychoacoustics

The perceived loudness of a sound depends on different factors not only on intensity.

The unit of measure for loudness is Phon

The acoustic pressure p is derived from the velocity v of the particles. Intensity is the power per unit area $I \sim p^2$. The scale is the decibel.

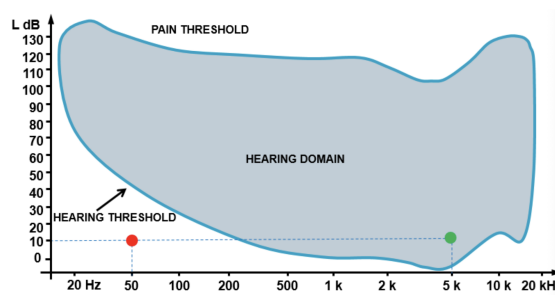
We define the base reference intensity as the minimum sound intensity that can be heard by a normal hearing person.

$$I_0 = 10^{-12} \left[\frac{W}{m^2} \right] \quad p_0 = 2 \cdot 10^{-5} [Pa]$$

$$dB = 10 \log_{10} \left(\frac{I}{I_0} \right) = 20 \log_{10} \left(\frac{p}{p_0} \right)$$

Doubling intensity means +3dB, doubling pressure is +6 db since $\log(2) \approx 0.3$

We do not hear all sound equally however:



Example

At 10 dB we hear 5kHz but not 50 Hz. A curve was empirically derived.

Also the **equal loudness curve** was defined, that **given a frequency tells you what pressure it should have to match the pressure of a 1 kHz sinusoid with 1000 pressure**. This is the **Phon scale**

The **sone scale** is used to describe relationships with sound of different loudness values. **A doubling of loudness is 10 phon.**

$$phon = 40 + 10 \log_2(sone) \quad sone = 2^{\frac{phon}{10} - 4}$$

Volley Theory: groups of neurons of the auditory system respond to a sound by firing action potentials slightly out of phase with one another so that, when combined, a greater frequency of sound can be encoded and sent to the brain to be analyzed

Localization Theory: pitch changes based on where maximum is in membrane

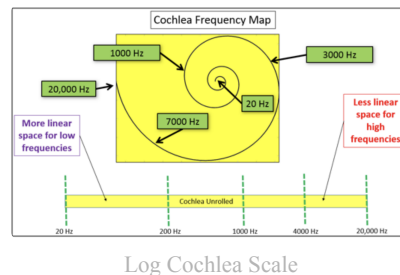
Temporal theory: pitch depends on temporal pattern of firing

One octave is double the frequency, there are 12 semitones:

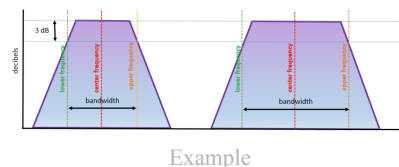
$$s^{12} = 2 \quad s = \sqrt[12]{2}$$

But our brain loses to distinguish the double frequency as an octave at high frequencies and needs more frequency

Critical bands are used to quantify the ability of the human ear to distinguish between individual frequency tones. The human ear can hear from 20 to 20,000 Hertz, but the ability to distinguish individual tones varies as a function of frequency.



This is given from the **cochlea** as it has a logarithmic spiral shape. The tones are therefore not distinguished but perceived as modulating or beating. These bands decrease in bandwidth as the intensity increases.



These bands are described via the **bark** scale. That is the distance of 1 CB is 1 bark.

Masking happens when a sound of higher intensity masks a sound of lower intensity. A pure tone can be masked with:

- +17 dB white noise up to 1 kHz
 - linear grow after 1kHz
- It can also be masked with another pure tone

The fundamental defines the pitch, the partials define the timbre.

1.4) Take Home Concepts

- **Basilar membrane** (auditory filters, tonotopic mapping):
 - **Auditory Filters:** The basilar membrane is narrow and rigid at its base and wide and soft at its apex, which allows it to act as a bank of filters tuned to different frequency bands.
 - **Tonotopic Mapping:** This spatial arrangement creates a mapping where high frequencies produce maximum vibrations at the base and low frequencies produce them at the apex.
- **Nonlinearities in the BM** (difference tones, Tartini's 3rd sound)
 - **Difference Tones:** When two sine tones are played loudly enough, the ear's nonlinear distortion creates additional "difference tones" at frequencies like $(x - y)$ or $(2y - x)$.
 - **Tartini's 3rd Sound:** This specific phenomenon occurs when two notes (like a perfect fifth) are played; the listener perceives a third, lower note produced by intermodulation distortion in the ear.
- **Loudness perception** (threshold of hearing, equal-loudness curves, loudness scales – phons and sones)
 - **Threshold of Hearing:** This is the minimum sound pressure level required for a sound to be just audible, which varies significantly depending on the frequency of the stimulus.
 - **Equal-Loudness Curves:** These curves (ISO 226:2003) describe the physical intensity needed for sinusoids of different frequencies to be perceived as having the same loudness as a reference 1 kHz tone.
 - **Loudness Scales (Phons and Sones):** The **Phon** scale matches the perceived loudness of any tone to the decibel level of an equally loud 1 kHz sinusoid, while the **Sone** scale is a linear scale where a doubling of loudness

corresponds to an increase of 10 phons.

- **Pitch perception** (sinuoidal/harmonic/inharmonic sounds, pitch scales – mel)
 - **Sound Types (Sinusoidal/Harmonic/Inharmonic):** Pitch perception varies by source: pure sinusoids have a clear height, harmonic sounds can maintain a "virtual pitch" even if the fundamental frequency is missing, and inharmonic sounds (like bells) may have unclear or complex pitch sensations.
 - **Pitch Scales (Mel):** The **Mel** scale is a perceptual scale that maps physical frequency to "ratio pitch," where doubling the mel value corresponds to a perceived doubling of pitch.
- **Masking and critical bands** (masking thresholds critical bandwidth, auditory filterbank, bark scale)
 - **Masking Thresholds:** This is the sound pressure level a test signal must reach to become just audible in the presence of a louder "masking" signal.
 - **Critical Bandwidth:** The human ear processes sound through "hearing bands" created by the cochlea; if two tones fall within the same critical band, they are difficult to distinguish and may result in "beating" or "roughness"
 - **Auditory Filterbank:** This concept treats the ear as a series of overlapping filters that quantify our ability to distinguish individual frequencies.
 - **Bark Scale:** The Bark scale is a psychoacoustic scale ranging from 1 to 24 that represents these discrete critical bands across the entire audible frequency range.

1.5) Stuff to Know

- Definition of Period, Frequency, velocity, wavelength, Standing waves
- Description of the human ear
- Sound Intensity: Decibel, Phon, Sone.
- Pitch
- Critical band, beating.
- Masking

2) Sound Synthesis & Sound Design

2.1) Sound Analysis

Sound is analyzed in the frequency domain which is consistent on how the human ear perceives sound. The characteristics of the sound can be better seen (high/low frequencies, harmonic vs non harmonic partials). Moreover **some filters are efficient in frequency domain**

We use the Discrete Fourier Transform (DFT)

$$X(k) = \sum_{n=0}^{N-1} x(n)e^{-j\omega_k n}$$

In computers this is implemented via the Fast Fourier Transform (FFT)

From N samples in the time domain we get N samples in the frequency domain.

For better understanding see [DSP 2 > 5.3\) Fast Fourier Transform \(FFT\)](#).

Each value of the FFT is called **bin** and represents the amplitude of a frequency of f_s/n Hz where f_s is the sampling frequency and n the size of the FFT. Since the DFT is periodic only the first $0 - f_s/2$ bins have an utility.

If the FFT is bigger than the signal, zero padding is used.

Also the Short Time Fourier Transform (STFT) is used to analyze **time varying signals**. STFT has a compromise. N is the window length: **bigger N implies better frequency resolution but worse time resolution** this is called the **uncertainty principle**. Also here see [DSP 2 > 5.3\) Fast Fourier Transform \(FFT\)](#).

To get a frequency response of a room we can find the Room Impulse Response (RIR) by recording the sound that a balloon makes in the quiet room. Since the balloon can be considered as a dirac delta:

$$y[n] = (h * \delta)[n] = h[n]$$

2.2) Classification of Sound Models

Various classifications of sound exist based on specific needs

Classification Based on Model Structure

This is based on **how the sound was generated**

- **Direct Generation:** techniques based on one or more independent blocks and sum of the results: **Additive and granular synthesis**
- **Feed Forward Structure:** feed forward multi block structure (lot of post processing): All non linear models **subtractive synthesis, ring and frequency modulation synthesis**
- **Interacting Structure:** interacting multi block (feedback). Most physics based models

Classification Based on Cognitive Representation

This is based on **what the model aims to represent**

- **Signal Based Models:** Represents shape of wave, the acoustic perception **ignoring the underlying physical model**. The focus is on receiver and human ear
- **Physics Based Model:** Represents acoustic signal as by product of physical mechanism of the production of the sound. Focus on emitter, sound source.

Classification Based on Underlying Techniques

These are based on the **processing domain**.

- **Time Domain Models:** based on time characteristics
- **Frequency Domain Models:** based on spectral characteristics

Filters are **linear** transformations, Modulation is a **non linear** one.

The processing is either on **sample basis** or most commonly on **block processing**, that is, N samples analyzed at a time like

with FFT.

These can be done via **hardware or software**

2.3) Time Domain Model Synthesis

This is a signal based approach. A characteristic of a musical sound is its complex dynamic behavior. Depends on external control and internal structure and generating process

Basic Generators

This produces a simple signal from a given data. The basic ones are oscillators for periodic signals and noise generators for random signals.

TABLE LOOKUP OSCILLATOR

An oscillator generates a simple periodic waveform. Instead of calculating this waveform, rather a lookup table with equidistant samples of the sound to generate is used. This is then amplified $a[n]$. Therefore: $s[n] = a[n]f[n]$

This is low quality approach as it lacks timbre and more complex modulations

NOISE GENERATOR

Natural sounds are not perfectly periodic. They contain varying statistical properties therefore a noise generator is required. Noise generator produces a sequence of random sounds with however clear statistical properties (flat spectral density)

2.3.2) Time-Segment Based Models

Sampling (Wavetable) Synthesis

This technique consists is sampling real sounds and then processing and playing them.

Some easy processing is speeding up the sound which heightens the pitch. But this has a bad effect. It lacks **prosodic** rules, which is the ability to modify the sound on the go with dynamics, vibratos etc. This also implements some little distortion.

The control parameters are the amplitude $a[n]$ and the fundamental frequency $f[n]$. The numeric array presents only one period of the periodic signal.

Granular Synthesis

Granular synthesis consists in dividing sounds into **grains**; small windows of acoustic elements in time domain. The mixing of these grains can recreate a complex sound. This was born from tape music.

From real sounds the grains are obtained by taking slightly overlapping windows to allow for both reproduction of the sound and the extraction of grains.

When segments are not very short we call it **synthesis by time segment processing**.

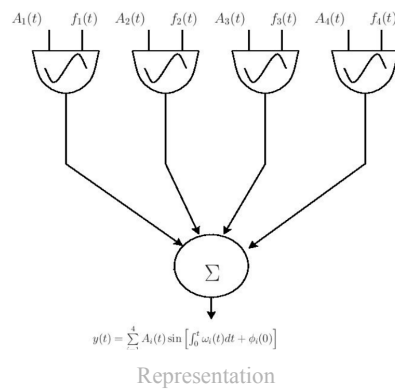
A second approach is the **micro-granular synthesis** which defines the grains as a windowed sinusoidal signal.

2.4) Frequency Domain Model Synthesis

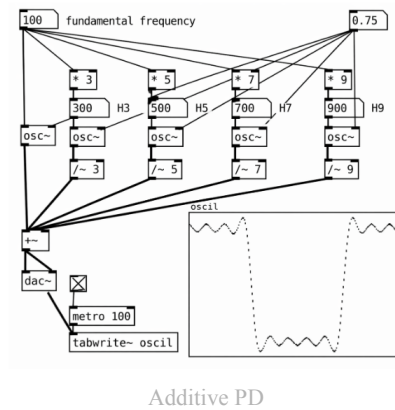
Additive Synthesis

Created by superimposing elementary sounds (sine waves). Any almost periodic sound can be decomposed into a series of simple sine waves. Additionally these waves can also change over time.

It is usually used when a specific sound is needed for example in psychoacoustic as it serves more as an analytical tool.



Additive synthesis can be done in PD.



We start with a base 100 Hz fundamental frequency. This is fed:

- directly to an oscillator
- to four odd integer multipliers

The odd integer multipliers show that only the odd harmonic partials are kept until $H9$. These are fed to an oscillator (with a gain of 0.75). Then the amplitude is divided by the partial index.

Finally all these signals are added together (additive synthesis) and are passed to a **Digital to Analog Converter (DAC)** in order to be played. The result is also printed in a table every 100 ms. We can see (and hear) that this approximates a square wave.

Wavetable vs Additive Synthesis:

	Wavetable	Additive
Principle	Plays back precomputed periods stored in tables	Builds sound by summing many elementary waves
Computational Cost	Low	High (many oscillators)
Control Over Spectrum	Indirect (post process)	Direct (modify directly the partials)
Dynamic Timbre Change	Through wavetable interpolation	modify partials
Sound Design Flexibility	Limited by available tables	Theoretically unlimited
Implementation Complexity	Simple	More complex to implement and control

3) Subtractive Synthesis

Subtractive synthesis consists in applying filters to a spectrally rich waveforms. Some frequencies will be subtracted and some enhanced. This does not suppose any kind of periodic signal.

3.1) Non Linear Model Synthesis

In a modulation we modify the og the signal (carrier) with another signal (modulator).

Amplitude Modulation

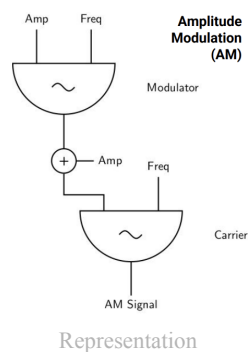
In AM we modify the amplitude and thus a multiplication is used:

$$x(t) = c(t)m(t)$$

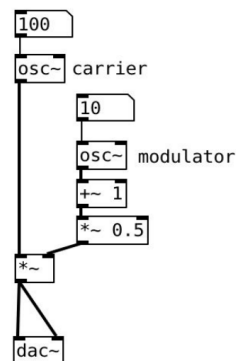
Some additional processing has to be done on sinusoids as they should not become negative (phase inversion, ring modulation).

$$x(t) = \sin(\omega_c t) \left[\frac{1 + \sin(\omega_m t)}{2} \right]$$

Some additional frequencies called sidebands are produced. These are the frequencies $f_{s1,2} = f_c \pm f_m$.



In PD this looks like, for example



PD Ring Modulation Example

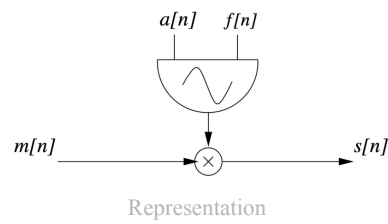
Ring Modulation

Ring modulation consists in multiplying the signal with a sinusoid carrier signal

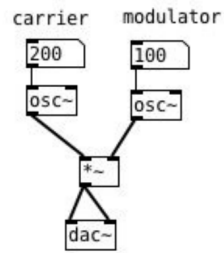
$$s = m[n] \cos(2\pi \frac{f_i}{F_s} n + \phi_c)$$

the produced output moves the spectrum in vicinity of the carrier

Only the sidebands $f_{s1,2} = f_c \pm f_m$ are present



In PD this becomes



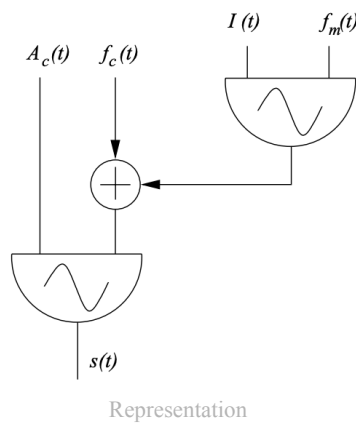
RM in PD Example

Frequency Modulation

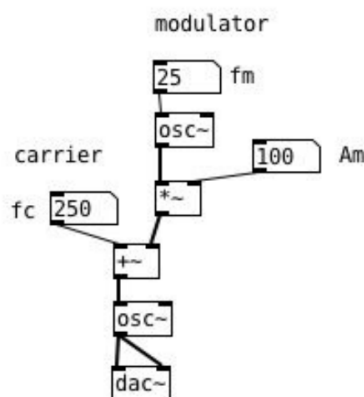
Here the frequency is modulated with another signal:

$$x(t) = \sin(\omega_c t + A_m \sin(\omega_m t))$$

where A_m is the frequency deviation which shows how much the frequency is changed.



In PD we have

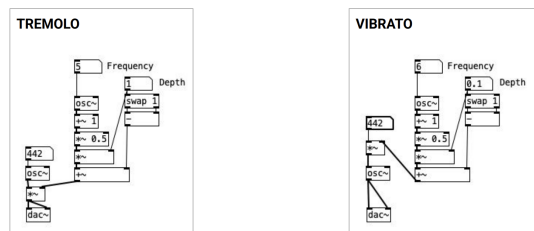


Frequency Modulation Example

3.2) Misc Stuff to know about subtractive synthesis.

Low Frequency Oscillator (LFO)

The **Low Frequency Oscillator (LFO)** is an oscillator with frequencies below the audible range (<20 Hz). It is used to modulate parameters. If higher frequencies are used, the result is AM or FM. The depth is the strength of the modulation

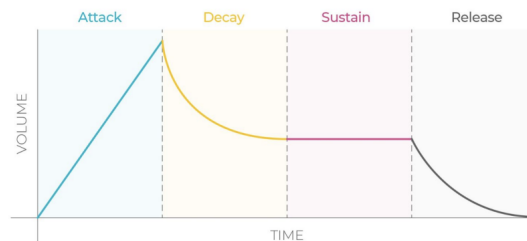


LFO Used for Tremolo and Vibrato

Amplitude Envelope

Each sound has 4 transients:

- **Attack:** start from zero to max value
- **Decay:** after max decrease
- **Sustain:** constant amplitude for a time frame
- **Release:** amplitude decreases to 0

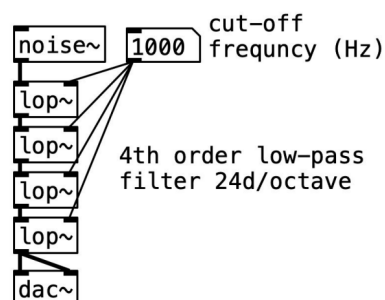


4 Phases of Sound

The evolution of a sound's amplitude over time, across its four transients, is called the amplitude envelope. In electronic music, it is often referred to by the acronym **ADSR (Attack, Decay, Sustain, Release)**.

Filters

A **Low Pass Filter (LPF)** cuts high frequencies after a threshold. The attenuation curve starts at the cutoff frequency and decreases to 0. In PD a LPF is the object **lop~**. The attenuation per octave is 6db per object, so concatenating them like in the image we can get more attenuation.



LPF in PD

4) Live Electronics TODO

5) Audio Archives Preservation

5.1) Historical Debate

Ongoing debate for 30+ years.

William Storm

1980: William Storm wrote first article on audio restoration and re-recording procedures with 2 common practices:

- Type I: Sound preservation + audio history
- Type II: Sound preservation of an artist

Type I is defined as:

The perpetuation of the sound of an original recording as it was initially reproduced and heard by the people of the era.

Type II is defined as:

The knowledge acquired through audiohistory preservation provides the sound engineer with a logical place to begin the next step—the search for the true sound of an artist.

Dietrich Schuller

Unesco: Unesco commissioned a guide on *save history, not rewrite it* influenced by **Dietrich Schuller**.

Analyse what the original carrier represents, technically and artistically, and to start from that analysis in defining what the various aims of re-recording may be.

Three approaches were finally formulated:

- **Type A:** same as Type I of Storm
- **Type B:** recording as it has been produced, precisely equalized for intentional recording equalizations, compensated for eventual errors caused by misaligned recording equipment and replayed on modern equipment to minimize replay distortions
- **Type C:** The recording as produced, but with additional compensation for recording imperfections caused by the recording technique of the time

Type B uses external info on the original recording to compensate for the imperfections. The same post processing (eq, filters, noise gates) are used. The reconstruction is more objective but still lenient to a slight subjectivity.

Type C is more in depth and must be documented as the decisions might deviate too much from the original recording. This fixes strict technical limitation

Type	Description	Goal
Type A	Equivalent to Storm's Type I.	Reconstruct the "exotic" historical listening experience.
Type B	The Preservation Standard. Corrects intentional EQ and technical alignment errors using modern gear.	Create an objective "master copy" based on scientific knowledge.
Type C	Active Restoration. Compensates for technical limits (removing hiss, rumble, or fixing frequency response).	Create a "clean" version; requires strict documentation because it is subjective.

George Brock-Nannestad

George Brock-Nannestad focused on pre 1925 acoustic recordings and applied reverse engineering to them. He wants to find the true sound from the recording we are hearing.

- Treat recording as an invertible math function
- tries to "undo" the physics of the 1920s equipment to reveal the actual sound that entered the room.

5.2) Operational Protocol

The idea is to keep a preservation copy of the og document: an organized data set that groups all the information represented by the source document.

1. Preparation of the Carrier

- Physical Documentation
 - Photographic Documentation
 - Scanned Images
 - Data validation
- Visual inspection
- Chemical analysis
- Optimization of the carrier

2. Signal Transfer

- Analysis of recording format
- System setup
 - Replay equipment
 - Remediation equipment (converter, acquisition software, monitoring, etc.)
- Monitoring
- Data validation
- Archival of source carrier

3. Data Processing and Archival

- Metadata extraction
- Completion of copy

The input of the remediation process is an audio document, and the expected output is its preservation copy—along with the source document ready to be stored again. The og copy has fundamental value that cannot be replaced by any copy.

5.3) Passive Preservation

here is a list of carriers (see later)

Carrier	Period	Composition	Characteristics
Cylinders	1886–1950s	Wax or Nitrocellulose	Early acoustic format; extremely fragile.
Coarse Groove (Shellac)	1887–1960	Mineral powders + Shellac binder	Brittle; breaks easily if dropped.
Instantaneous Discs	1930–1950s	Cellulose coating on Aluminum/Glass	High Risk. Used for one-off recordings; prone to peeling.
Vinyl (Microgroove)	1948–Today	PVC-polyacetate copolymer	More durable but sensitive to heat and bending.

The 4 major threats are:

- **Humidity (Most dangerous):** Hydrolisis causes discs to fall apart. Also fungus and mold are more present
- **Temperature:** above 40C makes vinyl warp. Accelerates chemical deterioration
- **Mechanical Deformation:** discs must be stored vertically.
- **Dust & Dirt:** make interference. Also fingerprints damage

5.4) Methodology

These are the steps to follow:

1. What should be preserved?

Based on risk (vinegar syndrome in tapes), demand and obsolete equipment that is disappearing

2. Physical Preparation

Before fixing, good documentation with: hq photos, video of carrier. Also visual inspection and delicate cleaning.

3. Capture Phase

Here transform soundwaves in bits. Use highest-end replay equipment and A/D converters. 96 kHz and 24-bit minimum. Must be human assisted

4. Preservation Copy

Unimedia data set. Audio as BWF. Visual evidence, metadata and description

5. Ethical Standpoint

Do not remove tape errors (hiss, scratch). his noise is "secondary information" that tells the story of how the document was transmitted. It is better to have a high-resolution file containing the noise than a "clean" file where historical data has been filtered out.

6. Data Integrity

One wrong bit can destroy the whole process. The protocol requires three types of verification (MD5, SHA-1, and CRC32) to ensure the file hasn't changed over time. At least **two copies** must exist at all times in different physical locations to avoid a single point of failure.

5.5) Carriers

Preservation of archives is essential as carriers degrade quickly while playback devices and knowledge are fast obsolete.

Traditional preservation paradigm: Preserve original

This paradigm however does not take into account the deterioration of the media and the machines used to play them

The digitization process was not considered a good solution for preservation, but it had only an access-oriented value. The rapid growth of computer sciences quickly led to the obsolescence of technologies, that had a short lifetime expectation. As digitization assumed more importance (better performance, bitrate, etc) and started to be considered a good way to store information.

In acoustic music (or speech recordings) the carrier only acts as a container of the artwork, not part of it.

In electroacoustic music however the carrier coincides with the artwork: the carrier is the witness of the process.

Preservation vs Restoration

Physical degradation of the carrier can be slowed down (but not stopped) with good preservation strategies.

Preservation consists in keeping the original acoustic information and carriers. There are two types of preservation

- Indirect: operating on the environment in which the audio document is stored.
- Direct: operating on the carrier itself with procedures for the stabilization of its physical conditions, without causing an alteration of the structure and the composition

Restoration is the process of recovering the original information from a degraded carrier, possibly through digitalization.

Incision Based Carriers

The Phonographic Disc is the first carrier standardized for audio preservation. This immediately led to problems (also future formats)

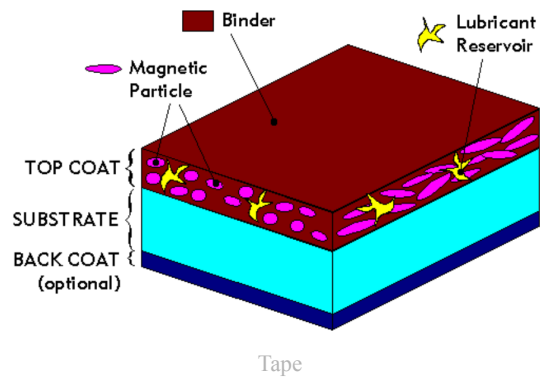
- From the standard, a disc should be run at 78 RPM. However due to different voltages (60 Hz vs 50Hz) the disc runs at slightly different speeds (78.26 vs 77.92 RPM)
- Disc is prone to degradation. The diamond needle slightly scratches the disc at every run so after a dozen listens the disc is damaged. Dust and scratches also damage the listening experience (crackle sound)
- They introduced a equalization. High frequencies are boosted, therefore an eq curve is needed to return a flat profile

Magnetic Based Carriers

These range from tapes to cassettes.

The first was the Open-Reel Tape (30's) and had various diameters and speeds.

In general the magnetic tape works as follows:



Vinegar Syndrome:

Humidity and temperature affect the acetic acid which degrades and damages the tape.

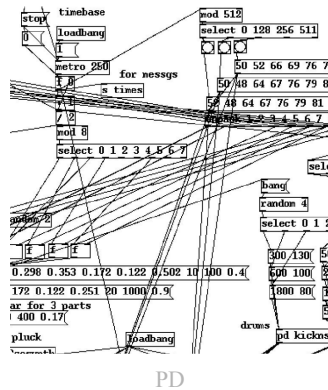
Soft Binder Syndrome– Sticky Shed Syndrome (SBS-SSS)

Deterioration of magnetic tape binder. The binder polymers used in magnetic tape constructions are subject to a **chemical process known as hydrolysis**. In this process, **long molecules are broken apart** by a reaction with water to produce shorter molecules. The **shorter molecules do not impart the same degree of integrity** to the binder system as do the longer molecules

This can be solved by baking (in the oven) at max 50-60°C for 1h up to 8 hours. However a wrongly baked tape can be irreparably damaged.

6) Pure Data

PD is a cross platform open source real time dataflow program environment



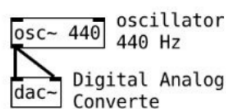
The creator is **Miller Puckette**.

There are 5 basic elements:

- **Objects:** specific class with inlets, outlets and methods. Can be data type or signal type
- **Messages:** contains numbers or symbols. The \$ symbol is a variable.
- **Number:** holds a number (can be read and can be written)
- **Comments:** comments
- **GUI:** gui elements like toggles, bangs

The two types of data are:

- **Data:** number/symbol
- **Audio Signal:** specific 32 bit float sequence. Sample rate and bit depth depend on sound card. Objects with tilde ~ work on audio signals (ex. osc~)

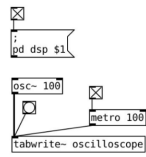


Osc DAC Combo

One of the fundamental blocks is the **osc~** and **dac~** combo. The osc generates a cos and the dac converts the digital signal into analog signal that we can listen to through our speakers or headphone.

The inputs of the oscillator are:

1. Frequency
 2. Phase [0, 1]
- How it is defined in the image, the frequency is already defined as 440.



Graphing Stuff

Arrays are defined via **tabwrite**. With the help of **metro** that sends a bang the array is saved (every 100 ms) and by activating the DSP this is printed to the GUI graph called *oscilloscope*

usually the inputs are **first hot, then cold**:

- **Hot:** object reads value of hot input path immediately
- **Cold:** object reads value of cold input path but doesn't update until the hot path is updated

7) Midi TODO

A synth is an instrument that generates audio signals. A sequencer is a device (or software) that can **manage and reproduce sequences of control signals**, which enable an electronic instrument to be controlled

To generalize these control signals a standard was designed: **Musical Instrument Design Interface (MIDI)**. It standardizes:

- digital interface and connectors
- communication protocol

MIDI 1.0 was proposed in 1983 and finalized in 1996.

Three hardware interfaces:

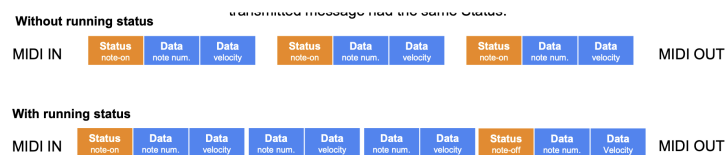
- **MIDI IN:** receives control signals
- **MIDI OUT:** sends generated signals
- **MIDI THRU:** replicates as MIDI OUT what was received as MIDI IN.

Not all ports are required on a single device.

A MIDI message is composed by **1 status byte** (first bit 1) followed by **one or more data bytes** (first byte 0)

- **Status Byte:** 1 TTT CCCC where T = type and C = channel
- **Data Byte:** 0 VVVVVVVV where V = value of data

If the same status is sent various times (with possible value changes) then only the data bytes have to be sent. This is called **Running Status**



Example

The most common messages are:

Tipo di messaggio	Codice (Status Byte / Tipo)	Data Bytes e relativi intervalli
Note Off	000	Data Byte 1: Note number (0-127) Data Byte 2: Velocity (0-127)
Note On	001	Data Byte 1: Note number (0-127) Data Byte 2: Velocity (0-127)
Polyphonic Key Pressure (Poly Pressure)	010	Data Byte 1: Note number (0-127) Data Byte 2: Pressure (0-127)
Control Change	011	Data Byte 1: Controller Voice (0-120 per Voice, 121-127 per Mode) Data Byte 2: Controller Value (0-127)
Program Change	100	Data Byte 1: Program/Preset Number (0-127) Data Byte 2: <i>Nessuno</i>
Channel Pressure (Mono Pressure)	101	Data Byte 1: Pressure (0-127) Data Byte 2: <i>Nessuno</i>
Pitch Bend	110	Data Byte 1: LSB - Valore Fine (range complessivo a 14 bit: 0-16383) Data Byte 2: MSB - Valore Coarse (Valore centrale di riposo: 8192)

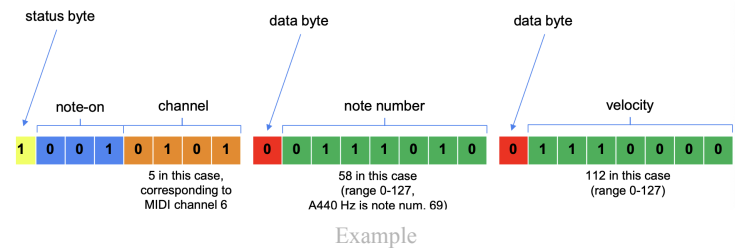
Here are some examples.

Note ON Message

Note ON: 001

The data is a note number (0-127) and then velocity (0-127)

Example:

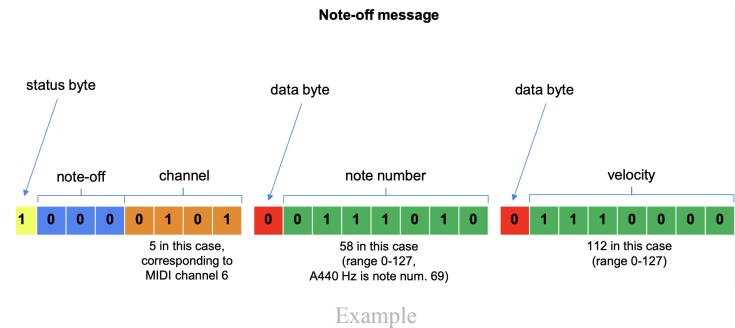


Note OFF Message

Note OFF: 000

The data is a note number (0-127) and then velocity (0-127)

Example:



8) Quiz

8.1) Acoustics and Psychoacoustics

Question	Possible Choices	Answer	Explanation
At which frequency is the human ear most sensitive among these values?	1. 20 Hz 2. 100 Hz 3. 2000 Hz 4. 20000 Hz	3	The human ear is most sensitive between 2000 and 5000 Hz, a range critical for speech. Extreme frequencies require higher intensity to be heard.
What is the term for the smallest change in frequency that can be detected by the human ear?	1. Pitch 2. Timbre 3. Just noticeable difference (JND) 4. Resonance	3	JND is the threshold where physical change becomes perceptible. Pitch is the perception of frequency, and timbre is sound quality.
What is the threshold of hearing for the average human?	1. 0 dB 2. 20 dB 3. 40 dB 4. 60 dB	1	0 dB is the standard reference for the minimum sound pressure audible to a healthy ear at 1000 Hz.
Frequency modulation in the cochlea is primarily used to:	1. Adjust pitch 2. Amplify soft sounds 3. Protect from damage 4. Encode frequency	4	The cochlea acts as a frequency analyzer, converting vibrations into neural signals to encode spectral info for the brain.
The basilar membrane in the cochlea is known for:	1. Constant thickness/stiffness 2. Incr. thickness/Decr. stiffness (apex to base) 3. Decr. thickness/Incr. stiffness (apex to base) 4. Uniform stiffness	3	The membrane is narrow/stiff at the base (high freq) and wide/flexible at the apex (low freq), allowing for tonal decomposition.
The term 'critical band' refers to:	1. Frequency group within which ear cannot distinguish tones 2. Middle ear response 3. Max amplitude tolerated 4. Smallest freq increment	1	If two tones fall within the same critical band, they interfere and the auditory system cannot resolve them individually.
The term 'tonotopy' refers to:	1. Intensity mapping 2. Spatial arrangement of frequency in cochlea 3. Temporal sequence 4. Neural synchronization	2	Tonotopy is the spatial organization where specific frequencies stimulate specific points along the cochlea and nervous system.
What is the role of the outer hair cells in the cochlea?	1. Transmit sound to brain 2. Act as microphones 3. Protect inner cells 4. Improve frequency resolution	4	They act as a "cochlear amplifier," physically changing shape to sharpen membrane vibration, improving sensitivity and selectivity.
In which medium does sound travel the fastest?	1. Air 2. Water 3. Steel 4. Vacuum	3	Sound travels fastest in solids due to high molecular rigidity. It cannot travel in a vacuum as there is no medium.
True or False: Sound waves are longitudinal waves.	1. True 2. False	1	In fluids like air, sound travels as longitudinal waves where particles oscillate parallel to the direction of wave propagation.
True or False: Higher frequencies have longer wavelengths.	1. True 2. False	2	Frequency and wavelength are inversely proportional ($v = f \cdot \lambda$); higher frequencies result in shorter wavelengths.
True or False: Frequency determines loudness.	1. True 2. False	2	Frequency determines pitch (high/low). Loudness is determined by the amplitude of the wave and sound pressure level.
True or False: Frequency determines speed.	1. True 2. False	2	Speed depends only on the physical properties of the medium (temperature, density). All frequencies travel at the same speed in one medium.
What is the speed of sound in dry air at 20°C?	1. 343 m/s 2. 434 m/s 3. 440 m/s 4. 3430 m/s	1	343 m/s is the standard value for air at room temperature. 440 Hz is a musical pitch reference, not a speed.

Question	Possible Choices	Answer	Explanation
What is the unit used to measure frequency?	1. Decibels (dB) 2. Hertz (Hz) 3. Pascals (Pa) 4. Newtons (N)	2	Hertz (Hz) measures cycles per second. dB measures intensity, Pa measures pressure, and N measures force.
What is the unit used to measure sound intensity level?	1. Decibels (dB) 2. Hertz (Hz) 3. Watts 4. Pascals (Pa)	1	Decibels are logarithmic units expressing the ratio of a physical quantity to a reference level, matching human perception.
What property of a sound wave determines its pitch?	1. Amplitude 2. Frequency 3. Wavelength 4. Speed	2	Pitch is the psychological perception of frequency; more cycles per second result in a "higher" perceived sound.
What property of a sound wave determines its timbre?	1. Amplitude 2. Frequency 3. Wavelength 4. Shape of the wave	4	Timbre is determined by the waveform's complexity (harmonics), allowing us to distinguish different instruments.
What type of wave is sound?	1. Transverse 2. Longitudinal 3. Electromagnetic 4. Mechanical	2	Sound is a mechanical wave, and "longitudinal" describes the particle motion being parallel to the path of the wave.

8.2) Sound Synthesis & Sound Design

Question	Possible Choices	Answer	Explanation
In what does granular synthesis divide the sound?	1. Frequencies 2. Waveforms 3. Tiny particles/fragments in time domain 4. Partial	3	Granular synthesis deconstructs sound into microscopic snippets (grains) in the time domain. It doesn't look at individual partials or abstract waveforms until reassembly.
What are physical sound models primarily concerned with?	1. Environmental effects 2. Simulating physical mechanisms 3. Perceptual aspects 4. Granular effects	2	Physical modeling focuses on the mathematical laws of physics governing how an object (like a string or tube) produces sound, rather than just mimicking the output wave.
What can nonlinear distortion and ring modulation change?	1. Frequency 2. Spectrum 3. Both frequency and spectrum 4. None	3	These processes create new frequencies (sidebands/harmonics) not present in the original signal, fundamentally altering both the frequency content and the harmonic spectrum.
What do hybrid models combine in sound generation?	1. Time domain models 2. Frequency domain models 3. Both time and frequency domain 4. None	3	Hybrid models leverage the advantages of both time-domain (like physical modeling) and frequency-domain (like additive synthesis) processing to create complex textures.
What does granular synthesis specifically divide sound into?	1. Partial	2	The "grain" is the fundamental unit of this synthesis. Partial
What involves separation into sinusoidal and noisy components?	1. Subtractive synthesis 2. Additive synthesis 3. Spectral Modeling Synthesis (SMS) 4. FM synthesis	3	SMS models sound as a deterministic part (sinusoids) and a stochastic part (residual noise). Additive synthesis only deals with the sinusoidal components.
What is the main focus of physical sound models?	1. Transformation 2. Synthesis 3. Simulation of physical mechanisms 4. Signal processing	3	While used for synthesis, the defining goal is simulating the mechanical or physical source of the sound (the "instrument") rather than the resulting signal.

Question	Possible Choices	Answer	Explanation
Which method is specifically used for voice compression?	1. Additive synthesis 2. FM synthesis 3. Subtractive synthesis 4. Linear Predictive Coding (LPC)	4	LPC models the human vocal tract. It is efficient for compression because it transmits the parameters of the vocal filter rather than the full audio waveform.
Which model emphasizes the shape of the sound wave reaching the ear?	1. Live electronics 2. Signal-based model 3. Frequency domain 4. Time domain	2	A signal-based model focuses purely on the resultant acoustic waveform and its characteristics as it arrives at the listener's ear.
Which synthesis method is complementary to additive synthesis?	1. Subtractive synthesis 2. FM synthesis 3. Granular synthesis 4. Spectral Modeling	1	Additive "builds" sound by adding sine waves (bottom-up), while subtractive "shapes" sound by filtering frequencies out of a complex source (top-down).

8.3) Subtractive Synthesis

Question	Possible Choices	Answer	Explanation
In subtractive synthesis, what does it mean when a filter is "fully open"?	1. It removes all frequencies 2. It allows all frequencies to pass through 3. It only allows the fundamental 4. It creates a resonant peak	2	A "fully open" filter (usually a low-pass) has its cutoff frequency set so high that it doesn't attenuate any audible harmonics, letting the raw signal pass.
In subtractive synthesis, what is often used to control parameters like filter cutoff or LFO rate?	1. Amplifier 2. Oscillator 3. Modulation wheel 4. Reverb unit	3	The mod wheel is a standard performance controller for real-time manipulation. Amplifiers and oscillators are internal sound modules, not tactile controllers.
What component is essential for shaping the timbre in subtractive synthesis?	1. Oscillator 2. Filter 3. Amplifier 4. Sequencer	2	The filter is the "subtractive" core; it shapes timbre by removing harmonics. The oscillator provides the source, and the amplifier controls volume.
What does ADSR stand for in the context of envelope generators?	1. Attack, Decay, Sustain, Release 2. Amplitude, Duration, Slope, Range 3. Adjust, Define, Shape, Refine 4. Audio, Distortion, Synthesis, Resonance	1	ADSR represents the four stages of a standard envelope: Attack (start), Decay (drop), Sustain (hold level), and Release (fade out).
What does the term "cutoff frequency" refer to in subtractive synthesis?	1. Highest oscillator frequency 2. The frequency where attenuation begins 3. Lowest oscillator frequency 4. Where the amplifier cuts the signal	2	The cutoff is the "border" frequency where the filter starts reducing the volume of the signal. It determines the brightness of the sound.
What does VCF stand for in the context of subtractive synthesis?	1. Very Complex Filter 2. Voltage Controlled Filter 3. Variable Cutoff Frequency 4. Vocal Chord Formant	2	VCF refers to a filter whose parameters (like cutoff) can be controlled by an external voltage, allowing for modulation by envelopes or LFOs.
What happens when you apply a low-pass filter to a sawtooth wave?	1. The sound becomes brighter 2. The sound becomes duller 3. The fundamental is	2	A low-pass filter removes high-frequency harmonics. Since sawtooth waves are rich in highs, removing them makes the sound "duller" or "warmer."

Question	Possible Choices	Answer	Explanation
	removed 4. Amplitude increases		
What happens when you increase the resonance of a filter?	1. Cutoff becomes lower 2. Frequencies around cutoff are emphasized 3. Overall volume decreases 4. Attack time becomes longer	2	Resonance (or Q) boosts the narrow band of frequencies right at the cutoff point, creating a "whistling" or "ringing" character.
What is the function of LFO in subtractive synthesis?	1. To generate the main audio signal 2. To modulate various parameters 3. To filter unwanted frequencies 4. To amplify final output	2	A Low-Frequency Oscillator (LFO) operates below the audible range to create repeating patterns like vibrato (pitch) or tremolo (volume).
What is the primary characteristic of subtractive synthesis?	1. Adding harmonics 2. Removing frequencies from a rich sound 3. Generating waves from scratch 4. Combining multiple oscillators	2	Subtractive synthesis starts with a complex, harmonic-rich wave (like a sawtooth) and "sculpts" it by filtering out unwanted parts.
What is the purpose of a noise generator?	1. Create pitched tones 2. Add unpitched, random frequencies 3. Control filter cutoff 4. Modulate amplitude	2	Noise generators produce random signals (white/pink noise) used for non-pitched sounds like wind, percussion, or adding "grit" to a tone.
What is the purpose of an envelope generator?	1. Generate initial waveform 2. Control how sound changes over time 3. Add harmonics 4. Mix different oscillators	2	Envelopes provide a "shape" to the sound, determining how volume, pitch, or filter cutoff evolves from the moment a key is pressed to when it's released.
What is the typical function of the sustain stage in an ADSR envelope?	1. How quickly sound reaches full volume 2. Level sound stays at while key is held 3. How quickly sound fades to silence 4. Initial volume level	2	Unlike Attack, Decay, and Release (which are time values), Sustain is a <i>level</i> value that sets the volume for as long as the note is held.
What type of synthesis is considered the opposite of subtractive?	1. FM synthesis 2. Additive synthesis 3. Wavetable synthesis 4. Granular synthesis	2	Subtractive removes frequencies from a complex source; Additive builds complex sounds by stacking simple sine waves together.
Which of the following is typically used as a sound source?	1. Sine wave 2. Triangle wave 3. Sawtooth wave 4. All of the above	4	Subtractive synthesis can use any of these; sawtooth and square are common for their rich harmonics, while sine/triangle provide purer foundations.
Which of these is an advantage of subtractive synthesis?	1. Perfect replication of acoustic instruments 2. Very little processing power 3. Highly intuitive and easy to understand 4. Generates infinite harmonics	3	Its popularity stems from its logical "source-then-filter" workflow, which is much easier for beginners to grasp than FM or Additive synthesis.
Which of these is NOT typically a core module in a subtractive synth?	1. Oscillator 2. Filter	4	Reverb is an effect often added at the end, but the core architecture of subtractive synthesis is defined by the VCO, VCF, and VCA chain.

Question	Possible Choices	Answer	Explanation
	3. Amplifier 4. Reverb		
Which parameter is typically NOT controlled by an envelope?	1. Volume 2. Filter cutoff 3. Oscillator pitch 4. Oscillator waveform	4	Waveform is usually a static setting. Volume, Filter, and Pitch are frequently modulated by envelopes to create dynamic expression.
Which of these waveforms is typically the richest in harmonics?	1. Sine wave 2. Triangle wave 3. Square wave 4. Sawtooth wave	4	A sawtooth wave contains all integer harmonics (1, 2, 3, 4...), making it the most complex and popular starting point for filtering.
Which type of filter is most commonly used in subtractive synthesis?	1. High-pass 2. Band-pass 3. Low-pass 4. Notch	3	The Low-Pass Filter (LPF) is the "classic" synthesizer sound, used to mimic the natural damping of frequencies found in acoustic instruments.

8.4) Audio Archives Preservation

Question	Possible Choices	Answer	Explanation
How does the methodology aim to minimize information loss in the preservation process?	1. By avoiding digital tools 2. Through specific re-recording recommendations 3. By ignoring technological advances 4. By using manual processes only	2	The methodology relies on precise technical standards (like 96kHz/24-bit) and specific signal chain setups to extract the maximum amount of data from the carrier.
What are the methodological principles designed to ensure in preserved audio documents?	1. Inaccuracy and unreliability 2. Accuracy and authenticity 3. Complexity and confusion 4. Minimal user access	2	By following a strict, documented protocol, the resulting "Preservation Copy" reaches a level of philological authenticity and technical accuracy.
What benefit does automation in the software provide in the preservation process?	1. Increases cognitive load 2. Reduces cognitive load during preservation 3. Decreases efficiency 4. Requires additional staff	2	Automation handles repetitive tasks (like metadata naming or checksum generation), allowing the operator to focus entirely on the delicate playback process.
What does automation in the software primarily reduce in the preservation process?	1. The accuracy of data 2. The cognitive load during the preservation process 3. The speed of processing 4. The reliability of preservation outcomes	2	Reducing cognitive load prevents human error during complex technical transfers, ensuring the operator can monitor the signal effectively.
What is considered integral to the methodology for preserving audio documents?	1. Financial considerations 2. Ethical considerations 3. Political implications 4. Geographic limitations	2	Ethics dictate the "Save History, Not Rewrite It" approach, ensuring the restorer respects the original document rather than altering it subjectively.
What is the design purpose of the software tools in the preservation process?	1. To complicate the preservation tasks 2. To automate and facilitate repetitive tasks 3. To increase the need for human intervention 4. To phase out all digital tools	2	Software is used to standardize the workflow, manage files, and validate data integrity without making the process unnecessarily complex.
Which areas have tested the effectiveness of the preservation methods described?	1. Asian art galleries 2. European audio archives 3. North American film studios 4. South American libraries	2	The protocol was refined and assessed through research projects involving some of the most prominent audio archives in Italy and Europe.

8.5) Live Music

Question	Possible Choices	Answer	Explanation
David Behrman's <i>Players with Circuits</i> (1966) involved:	1. Four performers 2. Entire orchestra 3. Two performers 4. A solo performer	1	This seminal work of early live electronics was scored for four performers using handmade electronic circuits.
From the 2000s onwards, personal computers became powerful enough to serve as:	1. Live audio patching and processing 2. Controlling 3. Both 1 and 2 4. None of the above	3	Modern PCs consolidated the ability to handle both the complex signal processing and the control data (MIDI/OSC) in one unit.
In Luigi Nono's <i>Quando stanno morendo</i> (1982), how many patch changes could occur in 7 seconds?	1. 3 2. 5 3. 7 4. 9	3	Nono's work with the SWR Experimentalstudio pushed the limits of analog/digital control, requiring up to 7 changes in a single bar.
In Nono's <i>Prometeo</i> , the "mobile sound" technique was:	1. Applied to voices and solo instruments 2. Used for spatialisation 3. A real-time synthesis technique 4. A live processing technique	1	The technique involved moving sound sources and processed signals through space, specifically applied to the acoustic performers.
In Salvatore Sciarrino's <i>Perseo e Andromeda</i> (1991), the 4i system was used for:	1. Sound spatialisation 2. Musical parts with freer timing 3. Playback of pre-recorded sounds 4. Both 2 and 3	2	The 4i system (developed by Giuseppe Di Giugno) allowed for more flexible temporal control than previous automated systems.
In Stockhausen's <i>Mikrophonie I</i> , the microphone was used:	1. Passively to record sound 2. Actively as a musical instrument 3. For amplification only 4. For spatialisation	2	Performers moved microphones physically across the surface of a large tam-tam, treating the mic as a "playable" excitation source.
In the 1970s, the biggest perceived problem in live electronics was:	1. Lack of power 2. Inability to vary patching during performance 3. Lack of interfaces 4. Lack of spatialisation	2	Hardwired analog patches were static; changing the "routing" of the signal mid-performance was nearly impossible until matrices and digital control arrived.
In the early years of live electronics, the sound director was often:	1. The composer 2. A technician 3. An electronic performer 4. Both 1 and 2	4	Before the role of "Computer Music Designer" was professionalized, the composer and a specialized technician usually shared these duties.
Stockhausen's <i>Mixtur</i> (1964) used how many ring modulators?	1. 2 2. 3 3. 4 4. 5	3	The orchestra was divided into 4 groups, each assigned to a ring modulator to be transformed in real-time.
Live electronics is primarily characterized by:	1. Live processing of voices/instruments 2. Synthesized sounds 3. Microphones/Amps only 4. Processing pre-recorded sounds	1	Unlike "tape music," live electronics focuses on the real-time transformation of acoustic sound as it is performed.
Max, developed by Miller Puckette in 1988, was initially:	1. Graphical environment for music control 2. Real-time audio software 3. Digital synthesizer 4. Spatialisation system	1	Max originally only handled control data (MIDI); the audio processing side (MSP) was added later in the late 90s.
Philippe Manoury's <i>Jupiter</i> (1987) was one of the first to use:	1. Score following and control 2. Real-time synthesis 3. Spatialisation 4. Graphical UI	1	It utilized "score following" (the computer "listened" to the flutist) to trigger electronic events at the correct musical time.
Steve Reich's <i>Pendulum Music</i> (1968) is an example of:	1. Live processing 2. Sonification through the Larsen effect	2	The piece relies entirely on microphones swinging over speakers, creating feedback pulses (the Larsen effect).

Question	Possible Choices	Answer	Explanation
	3. Synthesized sounds 4. Recorded sound parts		
The 4X system, developed at IRCAM, was:	1. Digital sound generation/processing machine 2. Analogue synthesizer 3. Spatialisation system 4. Control computer	1	The 4X was a groundbreaking, high-speed digital processor used for massive real-time transformations (e.g., in Boulez's <i>Répons</i>).
The combination "PC—MIDI—Sampler" led to the birth of:	1. Real-time computer music 2. 'Pseudo live processing' or 'studio electronics' 3. Digital synthesis 4. Analogue processing	2	It allowed performers to trigger high-quality studio-made sounds live, mimicking complex processing that was actually pre-recorded.
The Halaphon spatialiser was first used in:	1. Stockhausen's <i>Mixtur</i> 2. Nono's <i>Prometeo</i> 3. Halffter's <i>Planto por las Víctimas...</i> 4. Reich's <i>Pendulum Music</i>	3	Developed by Peter Lawo, the Halaphon allowed for sound rotation and movement, debuting in Cristóbal Halffter's work.
Main software for live electronics in the new millennium includes:	1. Ableton 2. PureData 3. Super Collider 4. CSound 5. Cubase 6. Audacity	2	While others are used, PureData (PD) is the open-source standard for real-time graphical programming in this context.
Zinovieff's EMS matrix made it easier to:	1. Edit the patch while playing 2. Control spatialisation 3. Vary parameters 4. Record sounds	1	By using a pin-matrix instead of patch cables, the user could instantly change signal routing during a performance.
The MIDI protocol, introduced in the mid-1980s:	1. Replaced analogue voltage control 2. Enabled digital audio processing 3. Allowed for real-time synthesis 4. Facilitated spatialisation	1	MIDI standardized the way digital devices talked to each other, replacing the cumbersome Control Voltage (CV/Gate) systems.
The Publison DHM 89 B2 (1978) offered:	1. Digital reverb 2. Pitch shifting (+1/-2 octaves) 3. Pitch shifting (+2/-1 octaves) 4. Both 1 and 2	4	This hardware unit was famous for combining high-quality digital delay/reverb with unique pitch-shifting capabilities.